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ENVIRONMENTAL STUDIES

ASSIGNMENT-2

Q.1 Describe the structure and functions of desert ecosystem.

Desert refers to a region or landscape in which the rainfall is negligible; and annual rainfall is less than 250 millimetres. they occupy about 17% of the earth's surface.

Deserts are characterised by

- (i) scanty flora and fauna
- (ii) hot days and cold nights
- (iii) soils with abundant nutrients but little or no organic matter

Structure and Functions :

(i) Abiotic components

Nutrients present in the soil and air are abiotic components. The organic substances are poorly present in the soil because of very low rainfall and high temperature.

(ii) Biotic components

Biotic components are producers, consumers & decomposers.

(a) Producers : In a desert, producers are mainly shrubs / bushes, some grasses & a few trees.

Example - Water retaining plants adapted to arid climate or soil conditions (succulents), hard grasses.

(b) Consumers : They include animals which are capable of living in xeric conditions.

Example - Insects, reptiles, some nocturnal rodents, birds & mammals like camels are also found.

(c) Decomposers : They are very few due to less vegetation and very low amount of dead organic matter.

Example - Bacteria & thermophilic bacteria.

Q2 Explain Biodiversity and its three level - Genetic, species and Ecosystem diversity

- Biodiversity is the diversity of plant and animal life in a particular region or in the world as a whole. It is often used as a measure of the health of biological systems.
- Living things contain in their cells, the basic instructions (which are often called genes) for their own development. Many of these instructions result in physical characteristics that affect the way organisms interact with their environment. Variations in such characteristics within the same species give rise to genetic diversity.
- species diversity is the measure of the diversity within an ecological community that incorporates the number of species in a community and the evenness of species abundances.
 - (a) species richness : no of species per unit area
 - (b) species evenness : evenness of individuals in a species
- Ecosystem diversity indicates the variation in structure and functions of ecosystems. It tells about trophic levels, energy flow, food and total stability of ecosystems.

Q3 Explain the terms - Extinct, endangered, vulnerable and rare species.

- When there is no reasonable doubt that the last individual has died, the species is said to be extinct. A wild species is extinct when exhaustive surveys in habitats have failed to record a single organism. - Example : Tasmanian Tiger, West African Black Rhinoceros
- A species is endangered if it is facing an extremely high risk of extinction in near future. Example : Mountain Gorilla, Amur leopard
- A species is vulnerable if it is not critically endangered or endangered but is facing a high risk of extinction in

near future. Example: Giant Panda, Polar Bear

- A rare species is a group of organisms that are very uncommon, scarce, or infrequently encountered.
Example: Nilgiri Tahr, Bengal Tigers, Asiatic Lion, Black Buck, Lion-tailed macaque

Q4 Describe the consumptive & productive values of biodiversity.

Consumptive and Productive Values

Fruits, vegetables, beverages and a wide array of food products are derived from biodiversity as they originate from living environment.

Medicines, fertilisers, fibres, rubbers and a number of other products obtained from or derived from the living environment have consumptive & productive values.

Q5 Discuss Human-Wildlife conflict.

Human-Wildlife conflict occurs when wildlife requirements overlap with those of humans, creating loss to both.

(i) consequences:

(a) wildlife conservation

(b) micro and macro economy

(c) safety and well-being for local population

(d) Food security

(ii) Reasons:

High human population growth rate which results in increasing demand for natural resources and the growing pressure for access to land.

Q6 What are the causes and consequences of Deforestation?

Causes:

(i) Population explosion

(ii) Agriculture: shifting cultivation, overgrazing, cash-crop economy

(iii) commercial logging: cutting trees for sale as timber or pulp

- (iv) Poverty
- (v) Mining
- (vi) Dams
- (vii) Infrastructure creation for logging
- (viii) Forest fires
- (ix) Acid rain
- (x) Development projects and housing projects

Effects:

- (i) Soil erosion: Soil is exposed to wind, sunlight, evaporation due to deforestation. Soil fertility goes down due to soil erosion & rapid leaching of essential mineral nutrients.
- (ii) Harm to Fisheries: As the soil is eroded, it accelerates siltation in dams, rivers and the coastal zone. The increased sedimentation harms downstream fisheries.
- (iii) More floods and droughts: Because of deforestation, there is no regulation of the flow into rivers. As a result, floods and droughts alternate in the affected areas.
- (iv) Habitat loss of wildlife: Butterflies, migratory birds & wild animals suffer due to loss of their habitat.
- (v) Extinction of some species: Many species are affected and some become extinct.
- (vi) Local and Global climate changes: The rainfall pattern is affected as the forest is cut down. Local & global climate changes may result from deforestation.
- (vii) Global warming: If the trees are burned, the carbon is released immediately as CO_2 which leads to global warming.
- (viii) Danger for the survival of local communities: Communities lose their source of food, fuel, construction materials and areas for livestock grazing by deforestation.

Q.7 Explain the consequences of over exploitation of surface water resources. What are the techniques for water harvesting for conservation?
Impacts of over-utilisation of underground & surface water:

- (i) Loss of Integrity of Freshwater Ecosystems
Human activities for infrastructure development like creation of dams, land conversion, etc are responsible for this loss of integrity of freshwater ecosystems. Water quality and quantity, fisheries, habitats etc. are at risk due to this loss of integrity.
- (ii) Risk to Ecosystem Functions
Population and consumption growth increases water abstraction and acquisition of cultivated land.
- (iii) Depletion of Living Resources & Biodiversity
Overharvesting and exploitation causes groundwater depletion, collapse of fisheries. Production of food, quality and quantity of water & supply of water gets badly affected by these depletions of living resources and biodiversity.
- (iv) Pollution of Water Bodies
Release of pollutants to land, air or water alters chemistry and ecology of water bodies. Greenhouse gas emissions produce significant changes in run-off and rainfall patterns. Because of water pollution, water supply, habitat, water quality, food production, climate change, etc, are at risk.

Measures to conserve water:

- a) Recharge groundwater by harvesting rainwater
- b) Use water wisely for household, agricultural and domestic purposes
- c) Reuse water whenever possible. For example, waste water after bath can be used for the toilet
- d) Avoid transmission and distribution losses by checking leaks in pipes, hoses, etc.

- (e) Prevent flow of untreated sewage to lakes & rivers .
This will reduce the likelihood of water pollution & help in water conservation.
- (f) collect water by building dams and reservoirs, & digging wells
- (g) Use drip irrigation, precision sprinklers for agriculture. Practice organic farming.
- (h) Adopt fairer policies for treatment, access & pricing of water.
- (i) Prevent flow of industrial effluents to natural water resources to avoid water pollution
- (j) Do protect forests to protect rivers, lakes, wells and other sources of water.